ChatGPT

Executive Summary

Building **Ksheera Setu** as an RFID-linked, offline-first dairy system for mastitis early-warning is a promising strategy that aligns closely with SIH26109's goals. Unlike prior work (e.g. GSSI et al. and others) that focus on classification of current mastitis from large datasets, our approach uniquely ties each milk measurement to an individual cow via RFID, collects data locally on an SD card, and emphasizes **personalized, time-series prediction** of mastitis 7–14 days in advance. Key steps include: attaching an RFID reader to identify each animal; using a milk sensor (EC, temperature, etc.) at milking; recording each cow's milk readings over time; immediately scoring milk quality (red/blue) for segregation; and uploading accumulated data for machine learning. The ML model will detect deviations from *each cow's own baseline* to forecast mastitis risk a week or two before symptoms.

This report analyzes SIH26109's requirements (7–14 day lead, multi-signal data, herd/dashboard alerts), reviews three relevant GitHub projects (summarized in the table below) and literature, identifies the gaps, and then details a **data collection plan**, ML pipeline, prototype architecture, validation plan, and next steps. All recommendations are grounded in public sources and prior art where available【3†L2171-L2174】【12†L207-L214】【57†L27-L34】.

SIH26109 Requirements and Evaluation

SIH26109 ("AI-Based Predictive Modelling for Early Forecasting of Bovine Mastitis in Indian Dairy Farms") is a hardware+software challenge asking for a system that *"predicts which animals are at risk one to two weeks before clinical signs appear"*, using data from farm sensors and records【57†L19-L24】【57†L27-L34】. The problem statement emphasizes a **7–14 day lead time**, integration of multiple signals (milk yield trends, conductivity, somatic cell count, lactation stage, parity, treatment/calving history, and environment) into *per-animal risk scores*, and dashboards or alerts for farmers and vets【57†L27-L34】【57†L38-L40】. The system should work at both animal and herd level, ranking animals by risk and explaining contributing factors.

No public Indian dataset meets these requirements; judges expect teams to clarify validation data or simulate realistic records【57†L75-L83】【57†L88-L92】. The SIHbuddy analysis notes that conductivity, SCC, and milk drop are well-known early indicators, so novelty lies in **temporal modeling and alerts**, not in which features to use【57†L98-L100】【57†L27-L34】. Importantly, the evaluation will scrutinize whether the claimed 7–14 day advance warning is genuine or simply re-stating assumptions, so we must justify how forecasts are validated (e.g. via CMT/SCC tests and veterinary records)【57†L75-L83】.

Repository Analysis

We reviewed three related GitHub projects: **gssi/mastitis-detection**, **cepdnaclk/Disease-Monitoring-in-dairy-industry**, and **FishMaster93/Cow_mastitis**. The table below compares their key aspects:

Project (Data)	Features Used	Label (Mastitis Defined)	Granularity / Data Size	Models & Performance	Code & Reuse Potential
gssi/mastitis-detection (Italy, ~41,104 records) 【12†L207-L214】	Multi-source features: monthly herd records including milk yield, fat/protein, somatic cell count (SCC), **electrical conductivity**, lactose, "functional check" (udder health), plus reproductive and demographic data 【12†L232-L239】 【12†L207-L214】. Builds **time-windowed** features from past 1–2 months.	**Clinical mastitis** diagnosed by veterinarian (from historical disease logs). Model predicts current mastitis.	~41,104 monthly records (Italian dairy farms) 【12†L207-L214】. Monthly granularity.	Ensemble tree classifiers compared (RF, XGBoost, LightGBM, CatBoost). Best: LightGBM. Accuracy ~88.3%, Precision 94.5%, Recall 81.6%, F1 87.6%【12†L207-L214】. Emphasis on explainability and feature reduction.	Mature: well-documented co with data-processing scripts. Contains dataset-assembly modules (e.g. `ele_conductivity_step.` `lactose_step.py`) and trained models. **Reusable for feature engineering and modeling code**【12†L258-L266】【12†L207-L214】.
cepdnaclk/Disease-Monitoring-in-dairy-industry (Sri Lanka university project)	Cow-level milk parameters: **lactation number**, milk **pH**, **electrical conductivity**, **fat%**, **salt content** 【23†L216-L224】. (Likely occasional or batch measurements entered via web app.)	**Subclinical mastitis (SCM)**. Model predicts CMT/SCC positivity from these milk parameters. Exact label not specified, but implied "SCM positivity" based on input features 【23†L216-L224】.	Small custom dataset (likely dozens of cows). Granularity: per sample (individual cow or bulk). No dataset size given.	Unclear – probably simple classifiers (e.g. logistic/ XGBoost). Focus is on web app UI. No performance reported. Likely baseline models.	Mostly front-end application. Limited code exposure. Some ML code under "ML Model" folder (we could attempt to inspect) but not easily accessible. **Low maturity** for heavy reuse, though the described features and workf (data input → ML → visualize are instructive.

Project (Data)	Features Used	Label (Mastitis Defined)	Granularity / Data Size	Models & Performance	Code & Reuse Potential
FishMaster93/ Cow_mastitis (Chinese dairy dataset from published J. Dairy Sci.)	Production and management data: **parity**, age (month), 305d milk yield, **peak yield/ day**, daily yield, fat%, protein%, **SCC (somatic cell score)** (used only to create label) 【80†L2360-L2369】. No sensor data.	**Clinical mastitis** defined via **somatic cell count/ score**: label = 1 if SCC ≥200,000 cells/mL or CMT score ≥2 (score≥4 in original Chinese, roughly SCC>200k) 【80†L2360-L2369】. Essentially uses high SCC as proxy for mastitis.	Dataset from one farm (size not given in code). Granularity: one "record" per cow per test (likely monthly). Models use ~9 different algorithms.	Compares 9 classifiers (RF, SVM, GBM, Logistic, LDA, NaiveBayes, AdaBoost, Decision Tree, MLP). Code suggests RF/XGB often top. Performance not explicitly given here, but J. Dairy Sci. paper likely reports AUC ~0.80+ (not cited directly here).	Scripts S1/S2 show full pipelir data loading (with translation 【80†L2344-L2352】【80†L23 L2369】, feature selection (dr SCC, etc.), and models. **Relatively complete** for classification pipeline. Can re data-cleaning and baseline model code, though dataset itself must be obtained separately (authors link Goog Drive).

These repos demonstrate that **binary classification** of mastitis from milk data is well-studied. The gssi and FishMaster projects show state-of-art classifiers and yields, while the cepdnaclk project shows a typical mobile/web interface for SCM prediction using conductivity and milk composition【23†L216-L224】. None, however, addresses *forecasting mastitis 7–14 days ahead*, nor ties individual animal IDs via RFID, nor uses an offline data-capture device.

Gaps vs SIH Requirements

All three projects provide useful insights, but they have key gaps relative to SIH26109's needs:

- **Lead-time forecasting**: None of the existing models predict several days or weeks before symptoms. GSSI and FishMaster simply detect current or recent mastitis from historical data; their labels mark existing infection. SIH requires *future risk estimation*. We must engineer features/trends (e.g. rising conductivity, yield drop) and validate on true future cases.
- **RFID-linked per-animal data**: The repos do not demonstrate collecting sensor data at the animal level in real time. They assume data are pre-merged by cow (often monthly records). Our RFID approach ensures each milk sample is matched to the specific cow's record, enabling true longitudinal tracking.

- **Offline data pipeline**: The challenge is hardware-oriented. We propose using an ESP32+SD-card to store data on-farm, unlike the repos which assume online data entry or preexisting datasets.
- **Validation (CMT/SCC)**: The SIH expects confirmation of predictions by tests like CMT or lab SCC. GSSI et al. mention "functional check" (udder exam) and possibly use clinical records【12†L230-L239】, but do not align predictions with actual next-week outcomes. FishMaster uses SCC as a proxy label【80†L2360-L2369】. We must plan direct comparisons: e.g. retest cows flagged high-risk with CMT/SCC in following week.
- **First-mile chilling/traceability**: SIH also has related hardware tasks, but the focus here is mastitis forecasting. We will include basic milk-quality segregation (red/blue can) for safety, but our core is the mastitis risk model.

In short, our project fills the niche of an **on-farm data capture + personalized time-series ML system**. We will integrate known mastitis indicators (conductivity, SCC, yield, etc.【57†L27-L34】【12†L232-L239】) in a new pipeline, and we must explain that we design for a 7–14 day forecast despite no off-the-shelf dataset【57†L75-L83】.

Data Collection & Labeling Plan

Sensors and Measurements: At minimum, each milking will use:

- **RFID Reader** to scan each animal's tag (key for data linkage).
- **Milk Sensor**: a device (e.g. Esp32 + probe) measuring **electrical conductivity (EC)** and **temperature** of milk【57†L27-L34】. Optionally, include: pH, fat content (if available from near-IR sensor), and volume (scale).
- **Ambient Sensors**: a small temperature/humidity sensor in the barn (to detect heat stress influences).
- **Activity Health Sensors (optional)**: if cows have pedometers or rumination monitors, integrate their data via manually keyed input or IoT.

Immediate Output (Red/Blue Segregation): The milk sensor will compute a simple "milk-quality score" (e.g. based on EC and temperature thresholds). If the score indicates abnormal milk (possible infection or contamination), milk is diverted to the **Red can**; else to **Blue can**. *This is for handling safety, not the same as the long-term forecast*. The bucket's cow ID, timestamp, EC, temp, and score are logged on the SD card.

Data Logging Schema: We will structure data as follows (for each milking event):

Column	Description
FarmerID	Unique ID of farm or owner
AnimalID	Cow/buffalo unique ID (from RFID)
Timestamp	Date/time of milking
AnimalType	Cow/Buffalo
Parity/LactNum	Lactation number or parity

Column	Description
Milk_EC	Milk electrical conductivity (mS/cm)
Milk_Temp	Milk temperature (°C)
[Optional: Fat%]	Milk fat % (if sensor available)
[Optional: pH]	Milk pH (if sensor available)
QualityScore	Immediate sensor score (normalized 0–100)
CanColor	`Blue` or `Red` based on quality score

Example CSV row (one milking):

```
FARM001,COW023,2026-09-25T06:15, Cow, 3, 5.8, 38.2, 78, Blue
```

("COW023" on its 3rd lactation produced milk with EC=5.8 and temp=38.2°C, scored 78/100, placed in Blue can.)

Additionally, we maintain master tables:

- **Farmer Table:** (FarmerID, Name, FarmLocation)
- **Animal Table:** (AnimalID, FarmerID, Type, Breed, Parity/LactationStartDate)

Records are appended for each milking. The ESP32 writes to SD card daily; at intervals (weekly/monthly), the card is uploaded to a central computer for analysis.

Labeling Mastitis Events: To train and validate forecasts, we must know when mastitis actually occurred. We plan:

- **Periodic Testing:** Perform weekly or bi-weekly **CMT (California Mastitis Test)** on each cow's quarters (a cow-side test indicating high somatic cells)【53†L116-L124】. CMT scores ≥2 (about ≥200k cells/mL) mark subclinical mastitis【53†L116-L124】. Record date and result per cow.
- **Veterinarian Diagnosis:** Log any clinical mastitis cases diagnosed by a vet (with date and cow ID) during the trial.
- **Milk Laboratory (if available):** Occasionally send milk samples for SCC to calibrate/confirm CMT and sensor readings.

These labels will create an "outcome" column in our data (e.g. Mastitis=1 if a clinical/CMT+ case occurred). We then link each prior-day measurement (timestamp) to whether that cow had mastitis diagnosed within the next 7–14 days.

Sample Size & Duration: Mastitis incidence varies, but to train a forecasting model we ideally want ~100+ events. If typical mastitis rates are ~20% annually per cow, then a herd of 100 cows observed for 6–12 months might yield 10–20 events. To be safe, we assume a pilot with **~200 animals over 12 months**, giving on the order of 200*0.2=40 events. More is better; if possible, run in multiple small farms or over multiple

seasons. Larger data reduces false positives. (No public constraints are given on farm size; we assume any typical cooperative dairy with 50–200 milking animals.)

Data Synchronization Workflow: Every day, the SD card's CSV is uploaded to a central server (or farm computer) via USB or wireless. ETL scripts will parse the new records, append to a database of milking data, and merge with animal/farm metadata. CMT/SCC results and vet logs are also imported. This builds our **longitudinal time series** for each animal. We assume a NoSQL or SQL database (e.g. a CSV-based ingest into TimescaleDB or Pandas).

ML Pipeline

Feature Engineering

We emphasize **temporal features relative to each animal's baseline**. From each animal's history, compute:

- **Rolling statistics** (e.g. 3-day and 7-day moving averages and trends) of milk EC, temperature, and yield (volume).
- **Deviation from historical mean**: For cow *i*, the difference between today's EC and cow *i*'s average EC over the past month, normalized by cow's std dev.
- **Yield drop**: If volume sensor is used, quantify sudden drop (e.g. ratio of last milking yield to 7-day avg).
- **Milk temperature**: A small increase (fever) could be a feature.
- **Parity / days in milk**: Time since calving, lactation stage.
- **Recent CMT history**: If previous week's CMT was borderline (score 1), include it.
- **Environmental factors**: If heat stress (high ambient temp/humidity) is present, it could elevate mastitis risk.

We will generate features similar to "time-windowed features" in GSSI's code【12†L232-L239】. For example, include EC over the last 5 milkings, linear trend slope of EC, and a "spike" indicator if EC jumps sharply. Likewise for yield and temperature.

These per-event features form the input `X`. The target `y` is binary: 1 if the cow shows mastitis (by CMT/clinical record) within 7–14 days after that event. (We may train separate models for 7-day and 14-day horizons.)

Models and Training

Candidate models: Gradient Boosting Machines (e.g. XGBoost or LightGBM), as used in GSSI【12†L207-L214】. These handle heterogeneous features and missing data well, and give feature importance. We will also try Random Forest and Logistic Regression as baselines, and potentially LSTM/GRU for sequence modeling if data volume allows. However, given limited on-farm data, tree ensembles are a safe choice.

We will train on historical events (many weeks of data). Because mastitis events are relatively infrequent, we'll address class imbalance (e.g. by oversampling or using appropriate loss). Cross-validation should be

time-aware: for instance, train on early months and test on later months, to mimic future prediction. We might also hold out entire cows as a "test herd" to ensure generalization.

Evaluation Metrics

- **Lead-time accuracy:** We need metrics for 7- and 14-day ahead forecasts. We will report precision, recall (sensitivity), and F1 for the binary prediction of "event in next 7/14 days". ROC-AUC is useful but hides class imbalance. A high recall (catching sick cows) with reasonable precision is critical; false positives (low precision) waste resources.
- **Calibration:** The model will output a risk score; we will check calibration (do 10% predicted risk events indeed have ~10% actual incidence?) via reliability curves.
- **Time-aware metrics:** We might compute the "average lead time" of detected events: among cows that actually had mastitis, what is the average number of days between alarm and diagnosis?
- **Comparative baseline:** As a strawman, compare against a simple rule (e.g. "flag if EC > threshold" or "if last CMT≥2"). Our ML must outperform these.

Explainability

Given the requirement that factors should be named for veterinarians, we will integrate SHAP or feature importance analysis. E.g. we can say "EC rose 20% and milk yield dropped 15% for Cow123, triggering high risk" (like SIHbuddy's "risk score climbing with conductivity and yield drift" example【57†L38-L40】). LightGBM/SHAP will allow us to show which features contributed to each risk score. This transparency is expected by SIH (it flags explainability as important【57†L128-L134】).

Prototype Architecture

Hardware Design

- **ESP32 microcontroller** with SD-card slot and RFID reader interface. Each milking station has this unit.
- **RFID Reader Module** (125 kHz or 13.56 MHz depending on tags) for scanning cow ear-tag when milking begins.
- **Milk Sensor Probe**: a waterproof conductivity and temperature probe (like those used in milk analyzers) inserted into the bucket. It measures EC and temperature continuously for a few seconds per milking and outputs values to the ESP32.
- **Power**: battery/solar or mains (depending on farm).
- **Data Storage**: SD card storing CSV logs as described.
- **LED/Display** (optional): to show scan success or sensor reading.
- **Red/Blue Can Setup**: a manual diverter; when sensor score is computed, a solenoid or manual lever directs milk to red or blue.
- **UI**: A simple LCD or the farmer's smartphone app can show "OK" or "FLAGGED" when scanning each cow.

Firmware logic (per milking): 1. Read RFID → identify AnimalID. 2. Dispense milk into bucket, activate sensor reading. 3. Record sensor values (EC, Temp) and compute an immediate score (e.g. linear scale of EC; CMT-like reagent analog not done). 4. Decide can: if score below threshold (good milk) → Blue; if above threshold

→ Red. 5. Append a CSV line on SD: [AnimalID, Type, Timestamp, EC, Temp, Score, CanColor]. 6. Go to next cow.

The **overall data flow** is summarized in the flowchart below:

```
flowchart LR
    A[Start Milking] --> B[Scan RFID (Cow ID)]
    B --> C[Collect Milk Sample]
    C --> D[Probe reads EC & Temp]
    D --> E[Compute Milk Quality Score]
    E --> F{Score > threshold?}
    F -->|No| G[Blue can: Accept milk]
    F -->|Yes| H[Red can: Flag milk]
    G --> I[Log data to SD card]
    H --> I
    I --> J[Next milking or End of Day]
    J --> K{SD card upload to Server?}
    K -->|No| A
    K -->|Yes| L[Upload CSV to PC]
    L --> M[Aggregate Data into Database]
    M --> N[Update Animal Time Series]
    N --> O[ML Model Run (daily/weekly)]
    O --> P[Risk Alerts + Dashboard]
    P --> Q[Dashboard/Email to Farmer/Vet]
```

In words: Each milking produces a logged record on SD (steps A–I). Periodically the SD card is uploaded and data ingested (L–N). A centralized script runs the ML forecasting model (O) to update each cow's 7–14-day risk. The dashboard then shows all cows with their current risk levels and histories (P–Q).

Sample SD-card CSV Schema

Below is a sample of what the CSV on the SD card might look like (comma-separated):

FarmerID	AnimalID	Type	Timestamp	EC (mS/cm)	Temp (°C)	Score	Can
F001	COW001	Cow	2026-09-25T06:00:00	4.2	37.5	85	Blue
F001	COW002	Cow	2026-09-25T06:05:00	7.1	38.2	40	Red

(Score is an internal QC metric; threshold 50 splits Blue/Red.)

Prototype Dashboard

The eventual dashboard (web or local app) would list each animal with columns: *AnimalID, Type, Latest Score, 7–14d Risk*. For example:

Animal	Type	Last Score	7–14d Risk	Action
COW001	Cow	85 (Blue)	Low	Normal
COW007	Cow	48 (Red)	High	Check Vet
BUF003	Buffalo	79 (Blue)	Moderate	Monitor

Clicking an animal reveals its history chart (EC over time, etc.) and a risk trend arrow (up/down).

Validation and Deployment Plan

Field Trial Design: We will pilot on one or two dairy farms (with farmers' consent). Each enrolled farm provides RFID tags and milking buckets for the sensors. We'll run the system for ~6–12 months to capture seasonal variation. The trial should include a veterinarian or trained personnel performing CMT tests (at least weekly) to label events. We record all antibiotic treatments and clinical diagnoses to compare against our alerts.

Ethical/Practical Issues: Farmers must consent to data use; data will be anonymized. The system only passively collects milk quality – it poses no extra burden aside from occasional CMT testing (a standard farm practice). False alarms (predicting mastitis that doesn't occur) could cause unnecessary worry or vet visits; we will emphasize the alerts are probabilistic and provide recommended actions (e.g. re-test via CMT rather than immediate treatment).

Risk & Limitations: Key risks include sensor malfunctions (we will calibrate EC sensors and test accuracy), data loss (SD redundancy/backups), and incorrect predictions. SIH analysis flags false positives leading to antibiotic misuse【57†L147-L150】; our recommendations will be conservative, e.g. *only alert, do follow-up CMT before treatment*. Also, smallholder farms may lack infrastructure (as [57] notes) – our offline SD approach helps, but farmer training on RFID scanning is essential.

We will iterate the model as data accrues, and continuously validate predictions against observed mastitis cases. Final success is measured by how early (days ahead) we can reliably forecast mastitis, and how this improves management (reduced mastitis incidence, less antibiotic use).

Prioritized Next Steps and Resources

1. **Prototype Hardware:** Build an ESP32 + RFID + sensor module. Use an affordable milk sensor (e.g. from DairyTech suppliers or DIY EC probe). Validate readings with known milk samples.
2. **Data Schema & Storage:** Implement the CSV format and test SD logging. Create a small test database (e.g. SQLite or TimescaleDB) to import from the CSV.
3. **Code Reuse:**
4. Leverage the *gssi* repository's code structure for data assembly. In particular, their `dataset_assembly` scripts【12†L258-L266】 show how to merge and transform raw sensor data. We can adapt `ele_conductivity_step.py` to process our EC readings into features.

5. Use GSSI's `ml_process/analyzer.py` workflow as a template for training/evaluating tree models. The LightGBM model they found best【12†L207-L214】 can be a starting point.
6. Borrow from FishMaster's preprocessing pipeline【80†L2344-L2352】【80†L2360-L2370】 for handling missing values and scaling (they impute with medians and standardize features). Their method of dropping "leaky" features (they drop SCC itself to prevent trivial prediction)【80†L2382-L2391】 is instructive if we want to simulate forecasting without immediate SCC input.
7. The cepdnaclk project's feature list【23†L216-L224】 reminds us to include basic milk constituents. If we ever add fat or pH sensors, their approach is relevant.
8. **Dataset Acquisition:** If possible, obtain any public datasets (the FishMaster Google Drive has their raw data; we can request access) to pre-train or validate. Also review academic datasets (e.g. a study with daily SCC readings).
9. **Mock Trials / Simulation:** Initially test models by holding out later data. We may simulate "lead time" by artificially shifting labels. The SIHbuddy suggests demonstrating on a known lactation where mastitis occurs【57†L38-L40】, so we could generate a synthetic example curve to demo.
10. **Dashboard Mockups:** Design UI screenshots or code (maybe reuse open-source farm dashboards) to show decision support (risk charts, animal list with risk color-coding).
11. **Stakeholder Engagement:** Identify a partner farm and local vet for pilot. Obtain ethical clearances if needed. Plan to log actual mastitis cases accurately.

We also note **SIH evaluation criteria**: solutions will be scored on technical merit, innovation, demo-ability, etc. Our *clear narrative* (RFID→sensor→ML) and modular architecture should satisfy **Clarity** and **Feasibility** (though SIHbuddy gave low feasibility due to data gap【57†L88-L92】, we will be transparent about training data). Emphasizing veterinary input and explainability addresses **Innovation Scope** (Green flag in [57]) and **Prototype** readiness.

Conclusion: By combining these elements – an offline IoT data pipeline, per-animal time-series ML, and a vet-backed validation strategy – Ksheera Setu can robustly meet SIH26109's requirements for 7–14 day mastitis risk forecasting.

Sources: Official SIH problem analysis【57†L19-L24】【57†L27-L34】; GitHub repositories (gssi【12†L232-L239】【12†L207-L214】, cepdnaclk【23†L216-L224】, FishMaster93【80†L2344-L2352】【80†L2360-L2369】); California Mastitis Test reference【53†L116-L124】.
